

Oct 24, 2025

Meeting Oct 24, 2025 at 15:41 CEST

Meeting records  Recording

Summary

Clément Lacaille led a session on Retrieval Augmented Generation (RAG), vectorization, and tokens, explaining how these technologies enhance LLM efficiency and data processing, with Naim Burrack, Sven Lück, and Sara Hofmann contributing to the understanding of tokens and vectors. The team then moved to a practical application, modifying an existing workflow to create a meal planning AI, with Leon Winde presenting the demonstration and Clément Lacaille guiding participants through the necessary steps. During the practical session, Clément Lacaille addressed technical issues encountered by Naim Burrack, Murphy Odion Usunobun, Chinedu Kirian Onwuzuruike, and Hana Abrham Tekle, providing detailed instructions on connecting to Pinecone and customizing system prompts for the meal planning AI. The meeting concluded with a discussion on future workflow sessions, where participants expressed preferences for individual or group projects, leading Clément Lacaille to offer both options.

Details

Notes Length: Standard

- **Introduction to RAG, Vectorization, and Tokens** Clément Lacaille initiated a discussion on Retrieval Augmented Generation (RAG), vectorization, and tokens, inviting participants to share their current understanding. Naim Burrack explained RAG as a method for LLMs to provide better answers by integrating information from external sources, and vectorization as the process of converting text into

numbers for computer processing. Clément confirmed Naim's understanding and elaborated on RAG's role in how LLMs efficiently retrieve vast amounts of data.

- **RAG Explained with a Restaurant Analogy** Clément used a restaurant analogy to explain RAG, comparing the LLM (like GPT) to a chef who quickly finds the best recipe (data) for a specific request without sifting through every option. This analogy highlighted RAG as a smart assistant that processes data quickly by understanding the request and matching it with exact data.
- **Vector Database and Vectorization Explained** Clément further elaborated on how RAG works internally, introducing vector databases and vectorization using a kitchen analogy. They explained that a vector database organizes data by "taste profile" (meaning) rather than by name, converting requests and data into numbers (vectors). This process involves dividing text into "tokens," which are then converted into numbers and grouped by category, allowing the system to search for meaning rather than exact words.
- **Efficiency Benefits of Vectorization** Clément emphasized that vectorization and tokenization significantly reduce processing time and cost for LLMs. By organizing data based on meaning, the AI can quickly retrieve relevant information, leading to faster response times and lower computational expenses compared to searching through all available data.
- **Tokens and Vectors in Detail** Clément demonstrated tokens as small chunks of input or output text that LLMs use to understand and generate language, providing examples like "apple" being split into multiple tokens. They clarified that vectors are numerical representations of these tokens, allowing the AI to match requests with data based on semantic meaning.
- **Participant Explanations of Tokens and Vectors** Naim described vectors as mathematical objects that represent the direction or meaning of words, while tokens are the small units (like words) that the language model uses as building blocks. Sven Lück added that tokens are the smallest units of text processed by algorithms, and vectors are lists of numbers representing meaning, which helps computers work with numbers instead of characters. Sara Hofmann offered a non-programmer's perspective, illustrating tokens as words in a sentence and vectors as the numerical description of the meaning. Clément summarized that AI converts questions into tokens, then into vectors (numbers), and searches its database for the closest meaning.

- Impact of RAG on Data Organization** Clément highlighted that RAG changes how data is organized for AI, moving away from structured formats like Excel to accommodate unstructured data like images, videos, audio, and large text files. This capability allows AI to process and remember context more effectively, making it central to various applications, including customer support and policy agents, by enabling direct interaction with models containing all necessary information.
- Practical Workflow Application** The meeting transitioned to a practical session to build a meal planning workflow, emphasizing that the previous day's work involved pushing data to the Pinecone vector database, which now holds Sven's preferences in a mathematical format. Leon Winde volunteered to present their screen to demonstrate updating the workflow.
- Workflow Modification Steps** Clément guided participants through modifying an existing Telegram voice agent workflow for meal planning. This involved duplicating the existing workflow, renaming it, and adding a "vector store question answer tool" within the AI agent node to access the meal planning data. The prompt for the AI agent and perplexity nodes was updated to reflect the new meal planning purpose, instructing the AI agent to use the vector database for personalized recommendations.
- Addressing Technical Issues and Confirming Steps** Naim and Murphy Odion Usunobun encountered difficulties in following the steps, prompting Clément to reiterate the process of duplicating the workflow and adding the vector store tool. Chinedu Kirian Onwuzuruike experienced connection issues, leading Clément to suggest restarting their system. Clément continued by instructing Leon to add an OpenAI chat model, specifically GPT-4, to the workflow.
- Connecting to Pinecone and Workflow Finalization** Clément guided participants to connect the vector store model to Pinecone, selecting "retrieve documents as a tool for AI agent" as the operation mode. They emphasized the importance of this connection for the AI agent to access the organized meal planning data, concluding with a request to download the updated workflow for further steps.
- Pinecone API and Document Retrieval** Clément Lacaille guided the group through setting up the Pinecone API, emphasizing that the "retrieve documents" option is crucial for the AI agent to pull relevant information from the Pinecone index. They explained that the "top four" or "top three" setting determines how many relevant matches the AI will retrieve from Pinecone based on a user's

request, with higher scores indicating better matches. Clément Lacaille also confirmed that including metadata is not required and is off by default, and clarified that the index in Pinecone is essentially a database for the stored data.

- **Embedding Models** Clément Lacaille explained that Open AI assists in converting user questions into tokens for processing, highlighting that the embedding model should always be "text embedding three small" for consistency with the server setup. Leon Winde inquired about the differences and costs associated with various models, to which Clément Lacaille responded that chunking and reimbursement methods vary across models. Naim Burrack suggested that Leon Winde use their own Open AI account instead of a shared one, but Leon Winde clarified that they were already using their own credentials.
- **Connecting Vector Store to AI Agent** Naim Burrack identified an issue where connecting the vector store to an AI agent consistently disconnected it for all users, explaining that the "make this tool for an AI agent" setting caused this behavior. Clément Lacaille acknowledged the problem and opted to proceed with the "answer question with vector store" option as an alternative.
- **Troubleshooting Connection Issues** Clément Lacaille assisted Hana Abrham Tekle with connecting the Pinecone vector store by guiding them to select the third option in "operation mode" and confirming their Pinecone account and index name. Hana Abrham Tekle successfully connected the vector store to the main flow and set up the OpenAI embeddings with the "text embedding three small" model, ensuring the use of a personal account.
- **System Prompt Customization for Publicity** Clément Lacaille initiated the customization of the system prompt for "Publicity" to tailor the AI's research capabilities. They decided that the AI should focus on meal ideas and recipes, including nutrition benefits, and conduct deep research from multiple sources, while avoiding source links. The group also decided to remove the instruction to "remove nutrition information" to ensure that nutrition details are included in the AI's responses, emphasizing the ease of modifying existing blocks for new use cases.
- **AI Agent Prompt Customization for Meal Planning** Clément Lacaille guided the team in writing the AI agent prompt, establishing that Publicity will provide options, and the AI agent will then use the Pinecone database to summarize data and create customized meal plans. They decided that the AI should provide two to three meal ideas with recipes that match user preferences, allowing for future

additions of dietary restrictions and other personal criteria. The group also agreed on a "short and conversational" tone for the AI's responses.

- **Workflow Setup and Verification** Clément Lacaille instructed everyone to copy the first Slack message into the "message publicity" system prompt and the second message into the AI agent system prompt to complete the setup. They then guided Murphy Odion Usunobun and Sara Hofmann through troubleshooting steps to ensure the correct messages were pasted in the system prompts and that all pins were removed from the nodes. Chinedu Kirian Onwuzuruike encountered more significant issues with their workflow, requiring them to watch the video or receive individual assistance later.
- **Testing the Meal Planning AI** Leon Winde demonstrated the AI's meal planning capabilities by asking for dinner ideas and receiving personalized suggestions based on past preferences. Clément Lacaille noted that the AI currently lacks memory of previous prompts and will need an additional memory button, which they plan to demonstrate in the following week. They also explained that the tool primarily looks into existing databases and can be connected to CRMs or Excel sheets.
- **Future Workflow Discussions** Clément Lacaille concluded the current workflow session, stating that the voice agent workflow was complete with two use cases: investment and meal planning. They opened a discussion about future sessions, proposing that participants work on individual or group projects, allowing for more ownership and practical learning. Clément Lacaille encouraged participants to share their preferences for future topics or to form groups to work on self-chosen projects, emphasizing that both individual and group approaches would be valid.
- **Feedback and Project Preferences** Naim Burrack and Sven Lück supported the idea of individual or group projects, highlighting the value of applying learned concepts to personal or professional scenarios. Several participants, including Claudia, Chinedu Kirian Onwuzuruike, and David, expressed a preference for continuing with a class-guided workflow, while others, like Naim Burrack, Navidid Zahed, and Sara Hofmann, opted for group projects. Clément Lacaille decided to offer both options, with specific guidance for each approach.
- **Looking Ahead to Next Week** Navidid Zahed, who missed the current session due to internet issues, expressed gratitude for the insights gained from previous weeks and looked forward to catching up and contributing ideas for next week's

projects. Clément Lacaille agreed to dedicate time on Monday to help Navidid Zahed and others who needed further assistance with the workflow. The meeting concluded with a discussion about forming groups for those who chose the project-based learning, with Clément Lacaille advising groups to have a maximum of three to four members.

Suggested next steps

- ☐ Leon Winde will make a duplicate of the Telegram voice agent workflow, call it meal planning workflow, and download the updated workflow, ensuring to remove the notes about the transcribe auditor video create a voice clone.
- ☐ Leon Winde will take a screenshot of the pine cone vector store tool configuration and download the updated workflow.
- ☐ Clément Lacaille will help Chinedu Kirian Onwuzuruike after the class with the workflow if he is not able to figure it out.
- ☐ Chinedu Kirian Onwuzuruike will watch the video after the class to resolve issues with their workflow setup, and if they are not able to figure it out, Clément Lacaille will help them on Monday.
- ☐ The group will think about their own project ideas or workflows over the weekend, with a deadline of tomorrow or Sunday morning, and post them in the Slack group.

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